Water at Mile High (x2) : Plant-Available Water in the Leadville Soil Series

How much Plant Available Water (PAW, in length units) is in the root zone of the Leadville soil series?

FSR Blog Post 1

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The Leadville series, subgroup Typic Cryoboralfs, can be found on mountain slopes, alluvial fans, and terraces along the center of Colorado and covers approximately 80,960 acres. The series is classified as loamy-skeletal, mixed, superactive Ustic Glossocryalfs. These soils are very deep and well drained, with substantial amounts of rock fragments throughout the soil profile (USDA NRCS, 2007).

For this analysis, I used primary characterization data from USDA Soil Survey Laboratory pedon 91CO037005, originally sampled as Leadville in Eagle County, Colorado, in 1991. I consulted my DRoCTS+ table as a reference, but based my calculation on the USDA Soil Survey Laboratory soil characterization data.

The series description for Leadville on SoilWeb identifies root-channel features in the B/E, Bt1, and Bt2 horizons, indicating that roots have penetrated below upper horizons. Because of this, I assumed a root zone extending through the A, E1, E2, Bt/E, and Bt1 horizons of the laboratory pedon, reaching a depth of 102 cm (approximately 40 inches).

The following calculations were performed for each horizon: (A horizon calculations below):

Available water fraction = (FC − PWP) ÷ 100

Soil mass per unit area = Horizon thickness × Bulk density

PAW mass per unit area = Soil mass per unit area × Available water fraction

PAW depth = PAW mass per unit area ÷ Water density

Water density was assumed to be 1.00 g/cm^3.

A horizon (0–18 cm):

Available water fraction:

36.1% − 8.5% = 27.6% = 0.276

Soil mass per unit area:

18 cm × 0.84 g/cm^3 = 15.12 g/cm^2

Mass of plant-available water:

15.12 g/cm^2 × 0.276 = 4.173 g/cm^2

Depth of plant-available water:

4.173 g/cm^2 ÷ 1.00 g/cm^3 = **4.17 cm PAW**

Table 1. Plant-available water calculated for five mineral horizons using USDA Soil Survey Laboratory data for pedon 91CO037005 using https://tinyurl.com/pawquestion57.

PAW Calculations by Horizon

Horizon	Thickness (m)	Bulk Density (g/cm^3)	FC (%)	PWP (%)	PAW Mass (g/cm^2)	PAW Depth (cm)
A	0.18	0.84	36.1	8.5	4.173	4.17
E1	0.20	1.61	13.7	1.3	3.993	3.99
E2	0.13	1.62	13.1	5.4	1.622	1.62
Bt/E	0.28	1.70	13.5	5.5	3.808	3.81
Bt1	0.23	1.69	15.8	6.2	3.732	3.73
Total	**1.02**				**17.327**	**17.33**

Total PAW = 17.327 g/cm^2 ÷ 1.00 g/cm^3 = 17.33 cm (6.82 inches)

PAW (%) = (17.33 cm ÷ 102 cm) × 100 = 16.99%

The estimated fine-earth plant-available water within the assumed 102 cm root zone is **17.33 cm, or 16.99% of the total root-zone thickness**.

The CDM/Welsh triangles estimates FC and PWP based on texture, but it does not take into account two very important soil attributes (hint: Rocks and Soil Organic Matter) that could change PAW. How would these two factors change the PAW in *your* soil?

The Leadville soil series contains large amounts of rock fragments. The series description identifies approximately 40% stones in the B/E horizon and 70% in the Bt1 horizon. Because rock fragments generally store much less plant-available water than fine soil particles (in-and-out water), ~17.33 cm is probably an overestimate. Organic matter could improve this by contributing to soil properties that retain more water. The Leadville series

has two shallow O horizons containing slightly decomposed needles, bark, twigs, and grass remains. These materials would help retain water in the soil.

Sources

California Soil Resource Lab. (n.d.). *Soil Data Explorer: Leadville – Water balance*. University of California, Davis. https://soilmap2-1.lawr.ucdavis.edu/sde/?series=leadville#water-balance

U.S. Department of Agriculture, Natural Resources Conservation Service. (2007). *Official series description: Leadville series*. https://soilseries.sc.egov.usda.gov/OSD_Docs/L/LEADVILLE.html

U.S. Department of Agriculture, Natural Resources Conservation Service, Soil Survey Laboratory. (1991). *Primary characterization data: Pedon 91CO037005, Eagle County, Colorado* [Laboratory characterization data]. National Soil Survey Center. https://ncsslabdatamart.sc.egov.usda.gov/